

MRT IN PRACTICE

Dual-Task Interference

Modality & Processing Code Effects on Task Performance

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Why This Matters in the Real World



WHERE MULTITASKING IS UNAVOIDABLE

Aviation

Pilots monitor instruments, comms, and flight path simultaneously

Driving

Navigation, hazards, and in-vehicle displays compete for attention

Control rooms

Operators track many streams of data under time pressure

Goal: find resource-sharing combinations that minimize interference.

Literature Review

Wickens (2024)

Multiple Resource Theory & Model

Introduces Multiple Resource Theory; explains how overlapping task resources increase interference and guides dual-task experiment design.

Kim et al. (2005)

Concurrent Working Memory Load

Shows working memory is divided by modality, supporting use of visual vs. auditory secondary tasks.

Leigh (1992)

Modality Congruence

Finds tasks using the same modality cause greater interference (modality congruence).

McPhee et al. (2022)

Task Similarity & Resource Competition

Demonstrates that task similarity and shared resources increase dual-task interference.

Jackson et al. (2023)

Dual-Task Interference Effects

Shows dual-tasking reduces performance in both tasks when cognitive demands overlap.

How do secondary task modality (visual vs. auditory) and processing code (spatial vs. verbal) affect primary performance, secondary performance, and workload during a visual-spatial tracking task in a piloting context?

Hypothesis Matrix (Two-Way ANOVA)

Effect	Null Hypothesis (H_0)	Alternative Hypothesis (H_a)
Modality	No effect on performance or workload	Affects performance and/or workload
Code	No effect on performance or workload	Affects performance and/or workload
Modality x Code	No interaction effect	Interaction affects performance and/or workload

Manipulations & Measures

INDEPENDENT VARIABLES

01 Modality

Visual

Participant sees the stimulus on screen

Auditory

Participant hears the stimulus through headphones

02 Processing Code

Spatial

Stimulus requires locating or responding to position in space

Verbal

Stimulus requires language-based identification

DEPENDENT VARIABLES

COGNITIVE MEASURES

01 Primary Task Performance

Mean cursor-to-target Euclidean distance. Logged automatically. Lower = better.

02 Secondary Task Accuracy

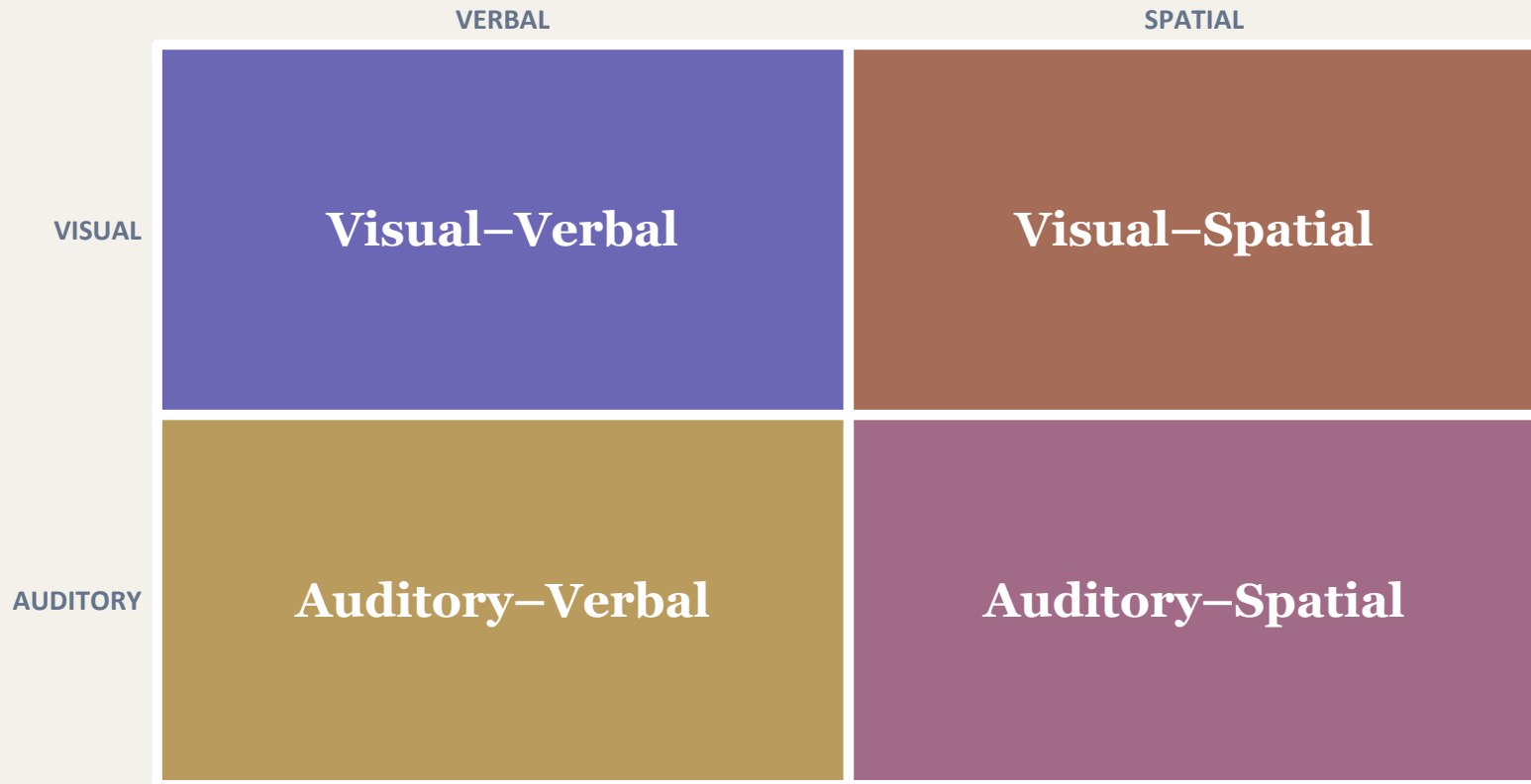
% of probes answered correctly via mouse click.

SUBJECTIVE MEASURE

03 NASA-TLX Workload Assessment

Self-reported workload (0–100) after each condition, split into Performance, Frustration, and Mental Demand.

Four Dual-Task Conditions



Primary task is always visual-spatial cursor tracking. The secondary task varies.



Example of Primary Task

Four Dual-Task Conditions

VERBAL

SPATIAL

VISUAL



Primary task is always visual-spatial cursor tracking. The secondary task varies.

Sample Size

$$n = 40$$

participants · within-subjects

WHY 40?

Statistical power

Enough to detect medium-sized effects in ANOVA

Normality threshold

$n \geq 30$ supports ANOVA assumptions

Dropout buffer

Room to lose trials without losing the study



Design Choices & Controls

Within-subjects. All 40 participants do all 4 conditions — more power per participant.

01 Counterbalancing

Order randomized per participant.

02 Primary task training

1-min practice flight before trials.

03 Clear task descriptions

Briefing + practice before each condition.

Click through to see each task →

Condition 1 of 4

Auditory-Spatial

PRIMARY TASK — ALWAYS ACTIVE

Keep the **white aircraft wings** aligned with the **gold flight director (V shape)**. Mouse = flight stick. Offset from center controls steering rate. Center = neutral. Off-screen = auto-centers.

THE DECISION — SAME EVERY TIME

You are flying over a warzone. Periodically you'll receive an alert telling you to either launch your **weapons** or deploy a **flare**.

L-CLICK	R-CLICK
Missile	Flare

The response is always left-click or right-click. Only how you *receive* the alert changes.

ALERT TYPE — DIRECTIONAL ALARM

You'll hear a buzzing alarm in your headphones. The **ear it plays in** tells you the answer:

Left ear	Right ear
= Missile	= Flare
→ Left-click	→ Right-click

Nothing on screen. Headphones required. The demand is localizing which ear the sound came from.

Trial lasts 25 seconds. Alerts every 3–5 sec. 3-second response window.

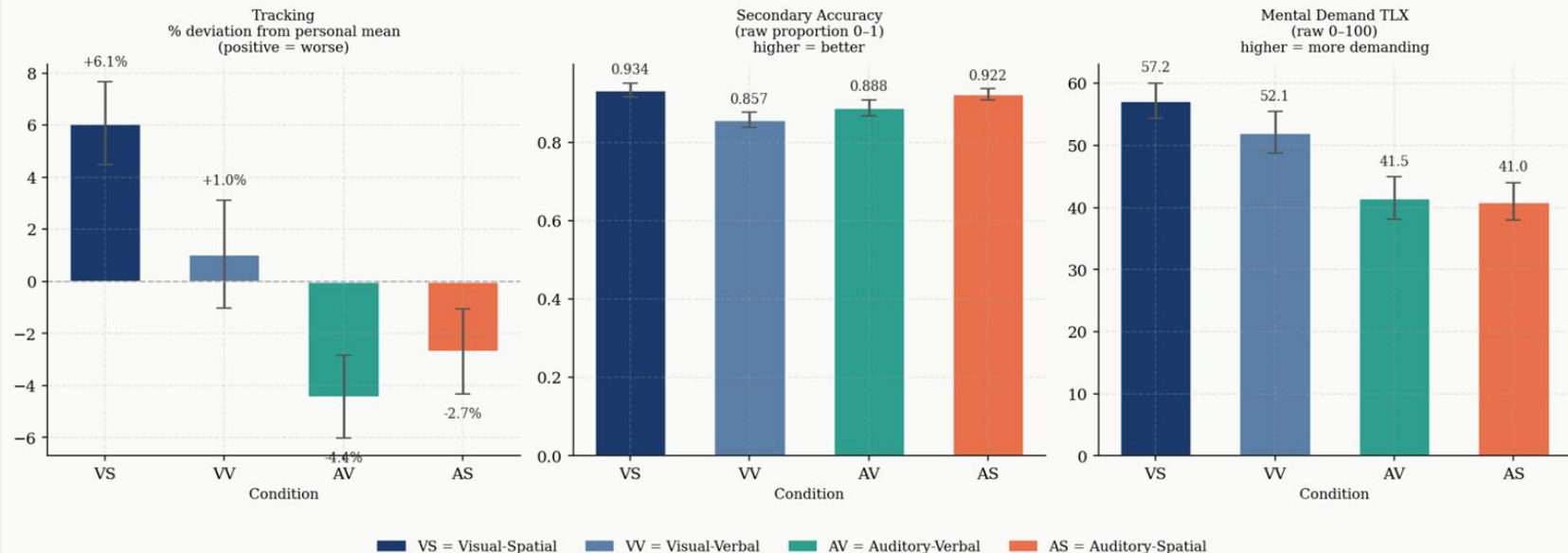
I'M READY — START TRIAL

2 x 2 Repeated Measures ANOVA

P-value: Probability our results are due to chance

F-value: How many times stronger our condition effect is compared to random noise

Modality	Code	Interaction
Tracking Error ($p = 0.04$, $F = 4.27$) Mental Demand ($p = 0.01$, $F = 8.04$)	Secondary Accuracy ($p = 0.01$, $F = 7.15$)	No significant interaction effects found

Figure 1 — Mean $\pm$ SEM by Condition

Primary Tasks

Audio Modality significantly better than visual modality

This supports the resource competition hypothesis

Secondary Tasks

Spatial code significantly better than Verbal code

This goes against the resource competition hypothesis

Mental Demand

Audio modality significantly better than visual modality

This supports the resource competition hypothesis

Figure 2 — Workload-Performance Alignment

Condition-level means — same rank order on both DVs

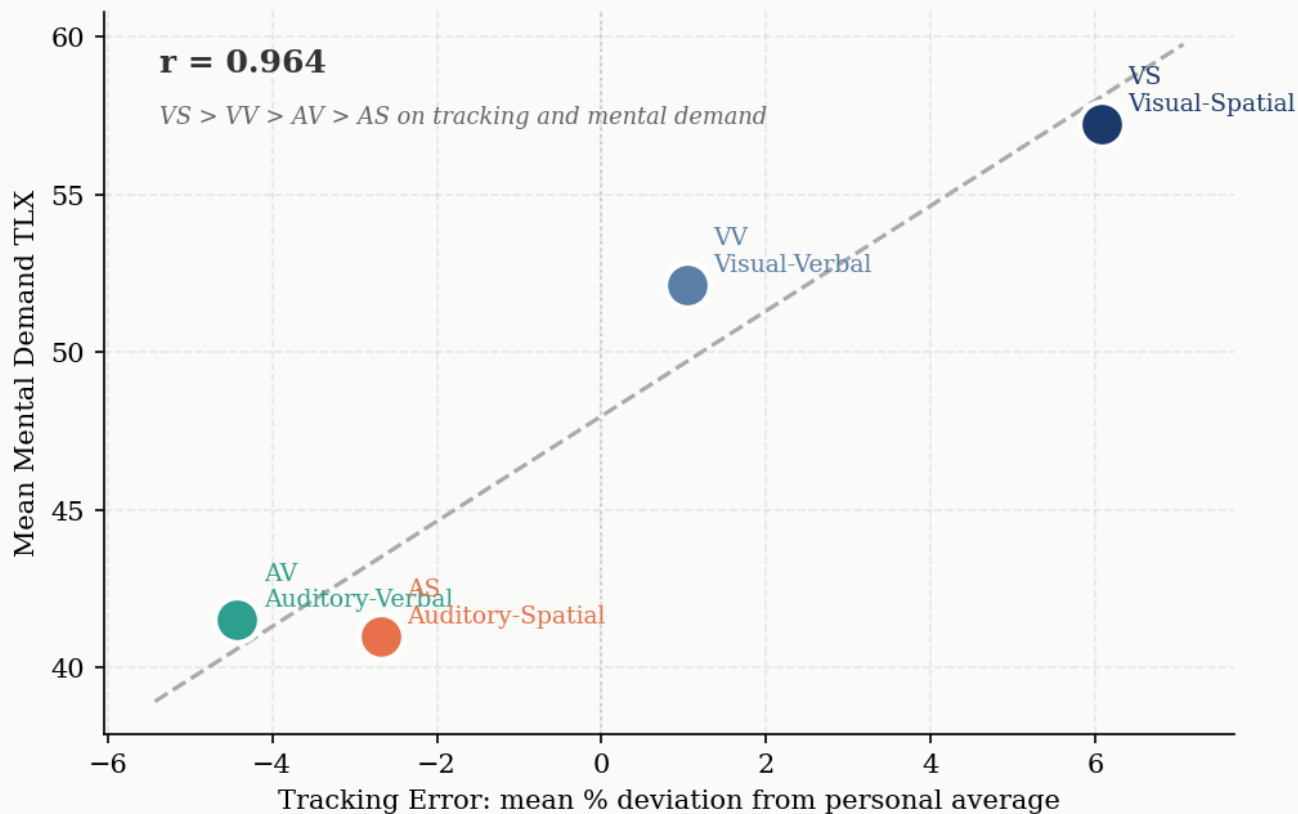
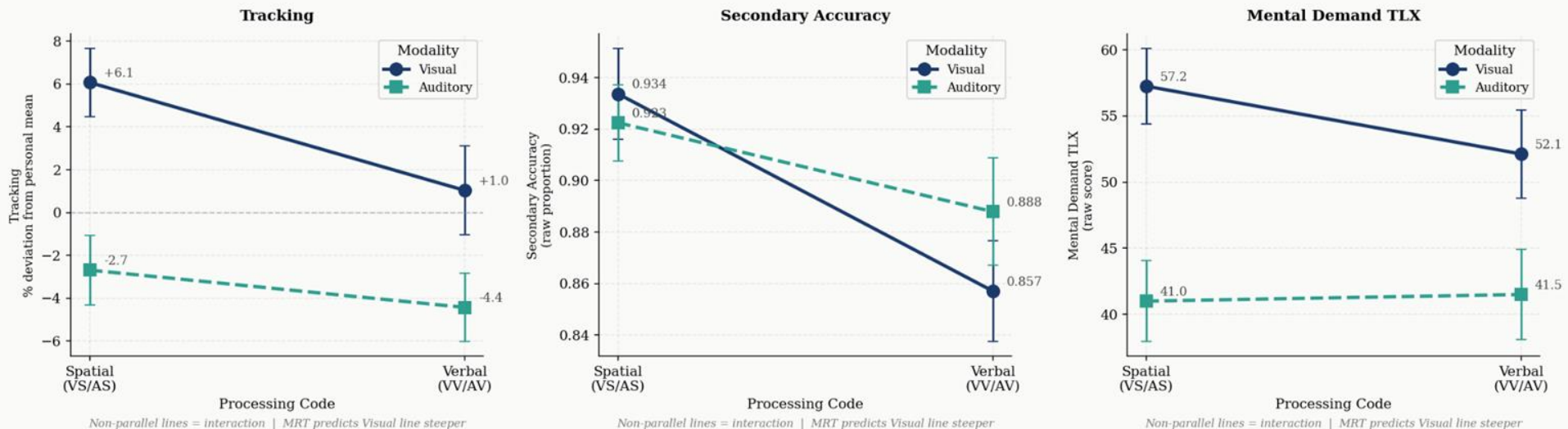


Figure 3 – 2×2 Interaction Plots: Modality × Processing Code



Limitations of Study Design & Future Work

Poor Design of Controls

Code Specific Control Schema

Designing a secondary-task control scheme that corresponds to people's existing mental models for the spacial tasks serves as a primary expected cause of improvements in secondary task performance for spacial code formats.



Unintuitive NASA-TLX Performance Scale

Increasing the performance slider means reduced performance.

Performance

How successful were you in accomplishing what you were asked to do?



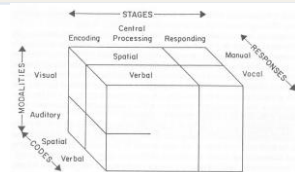
Inconsistency of study environment

Lack of Experimental Control

The means in which we administered the study was poorly controlled. Five separate individuals administered the tests each in separate environments and on different hardware.

Only 2 Axes Considered

Wickens' MRT model utilizes 4 axes



THANK YOU

— Questions & Discussion

Tracking Score Two-Way ANOVA		
	F-Value	p-value
Modality	4.2681	0.0455
Code	0.2830	0.5978
Modality x Code	0.6551	0.4232

Secondary Task Accuracy Two-Way ANOVA		
	F-Value	p-value
Modality	0.2972	0.5888
Code	7.1518	0.0109
Modality x Code	1.1761	0.2848

Frustration TLX Two-Way ANOVA			Performance TLX Two-Way ANOVA		
	F-Value	p-value		F-Value	p-value
Modality	1.9014	0.1758	Modality	0.1408	0.7096
Code	0.5478	0.4637	Code	0.4637	0.4999
Modality x Code	0.5244	0.4733	Modality x Code	0.0162	0.8995

Mental Demand TLX Two-Way ANOVA		
	F-Value	p-value
Modality	8.0439	0.0072
Code	0.1601	0.6912
Modality x Code	1.0924	0.3024

Figure 4 — DV Inter-Correlations

Participant-level averages across conditions

